

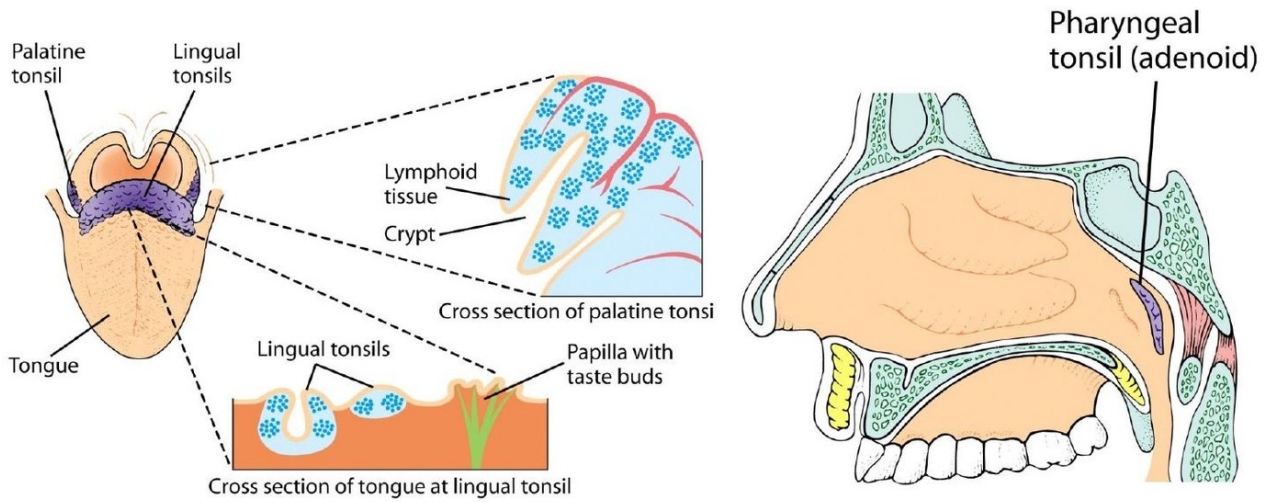
Immunology of the Gut

Companion Figures (Lecture 1, BRS 1b)

Figures referenced in the Notion revision file as ■ *Figure L1.X*. One figure per page, in the order they appear in the .md.

of venous (HEV) postcapillary venous
stream

The oral cavity is rich in immunological tissue



throughout the gastrointestinal tract and relate to local function

Figure L1.1 — Tonsils of the oral cavity (palatine, lingual, pharyngeal/adenoid). MALT exists throughout the GI tract, including the mouth.

Issue (GALT)

Is

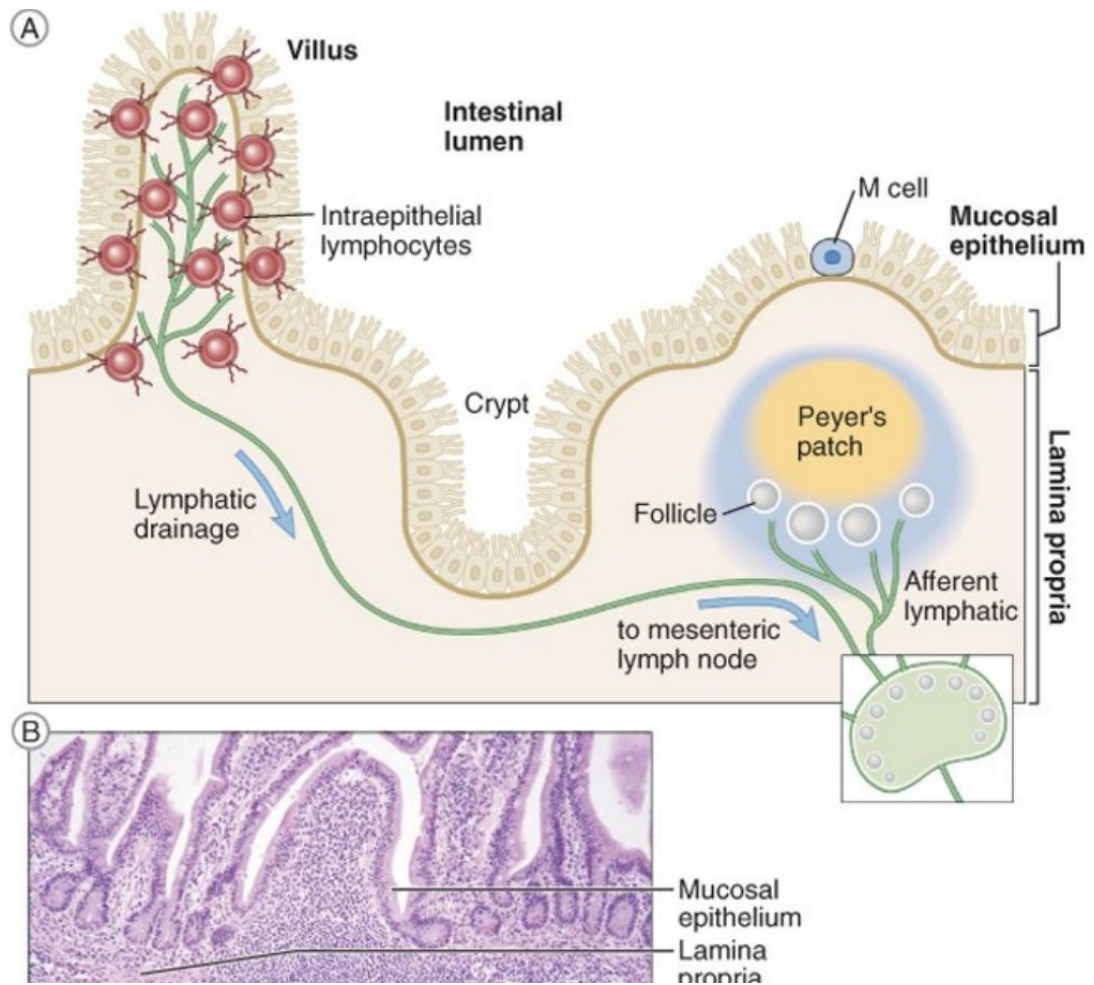
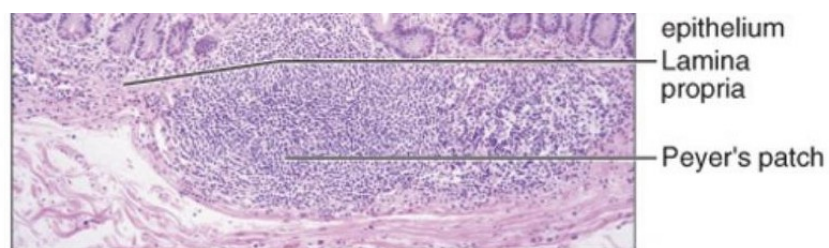


Figure L1.2 — GALT overview: villus with intra-epithelial lymphocytes, Peyer's patch beneath the M-cell-bearing epithelium, lymphatic drainage to the mesenteric lymph node.

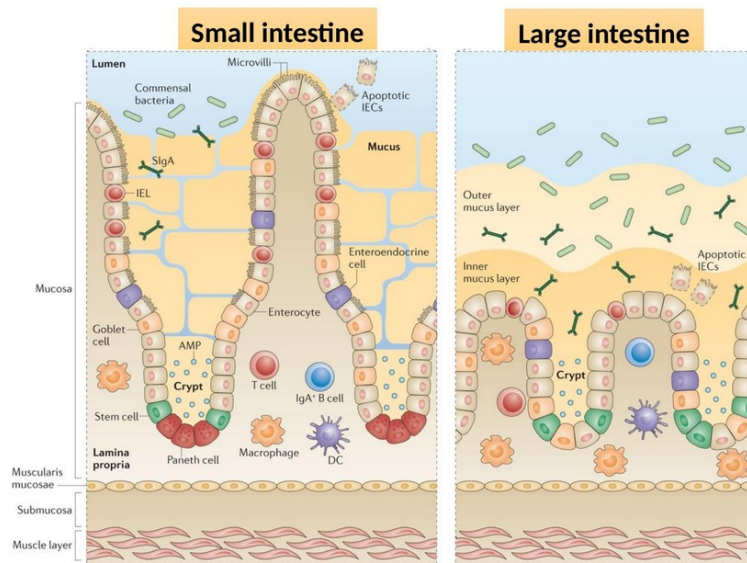
ent



tinal tract and relate to local function

maintaining systemic immune homeostasis

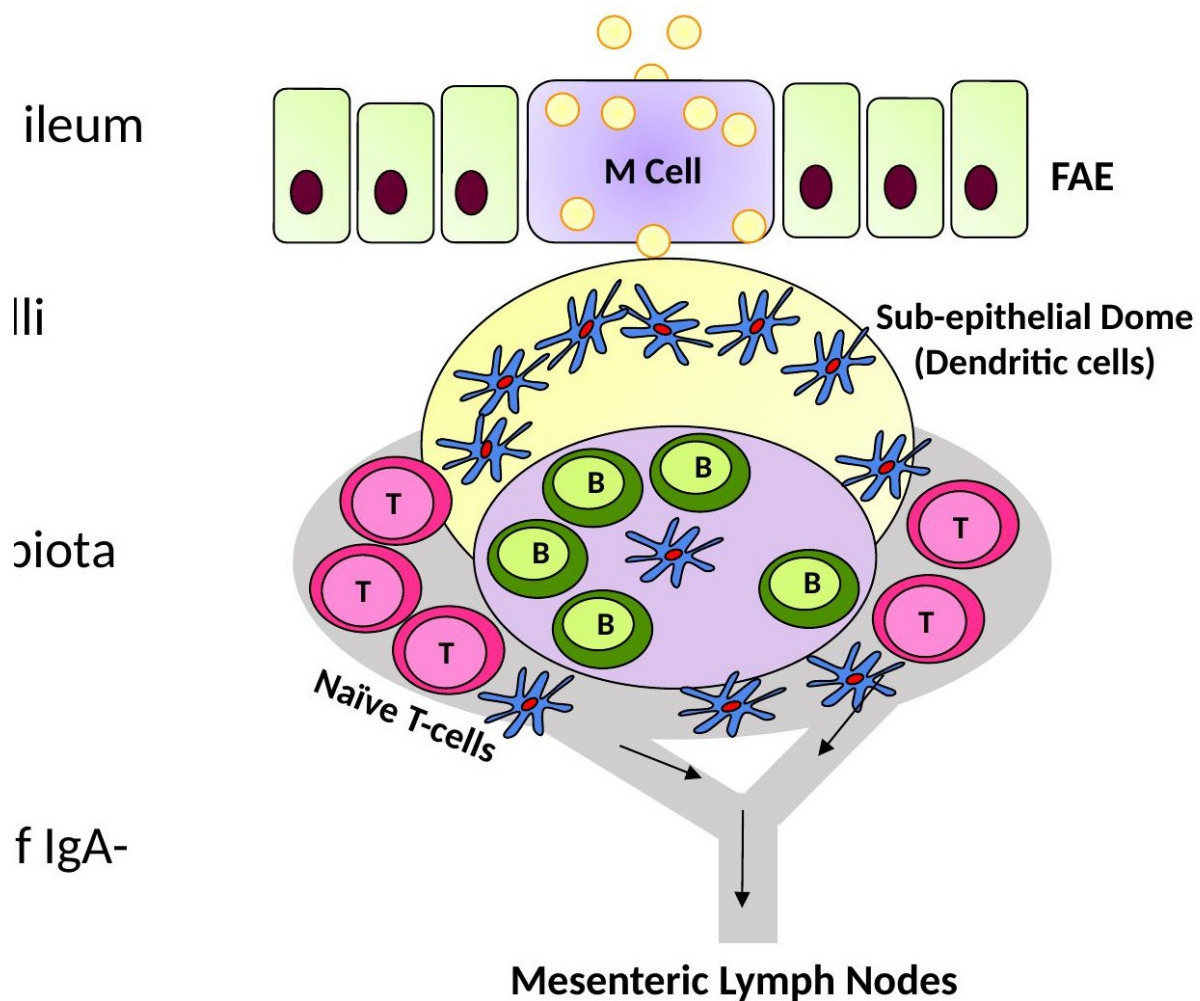
Figure L1.3 — Peyer's patch on H&E;; mucosal epithelium overlying lamina propria with a dense lymphoid aggregate (the patch itself).



the gastrointestinal tract: Describe the cellular arrangement of tissues throughout the gastrointestinal tract and relate to local function

Figure L1.4 — Small vs large intestine cellular cast. Small intestine: villi + crypts, Paneth cells at crypt bases, single mucus layer. Large intestine: crypts only (no villi), NO Paneth cells, double mucus layer (outer with bacteria, inner sterile).

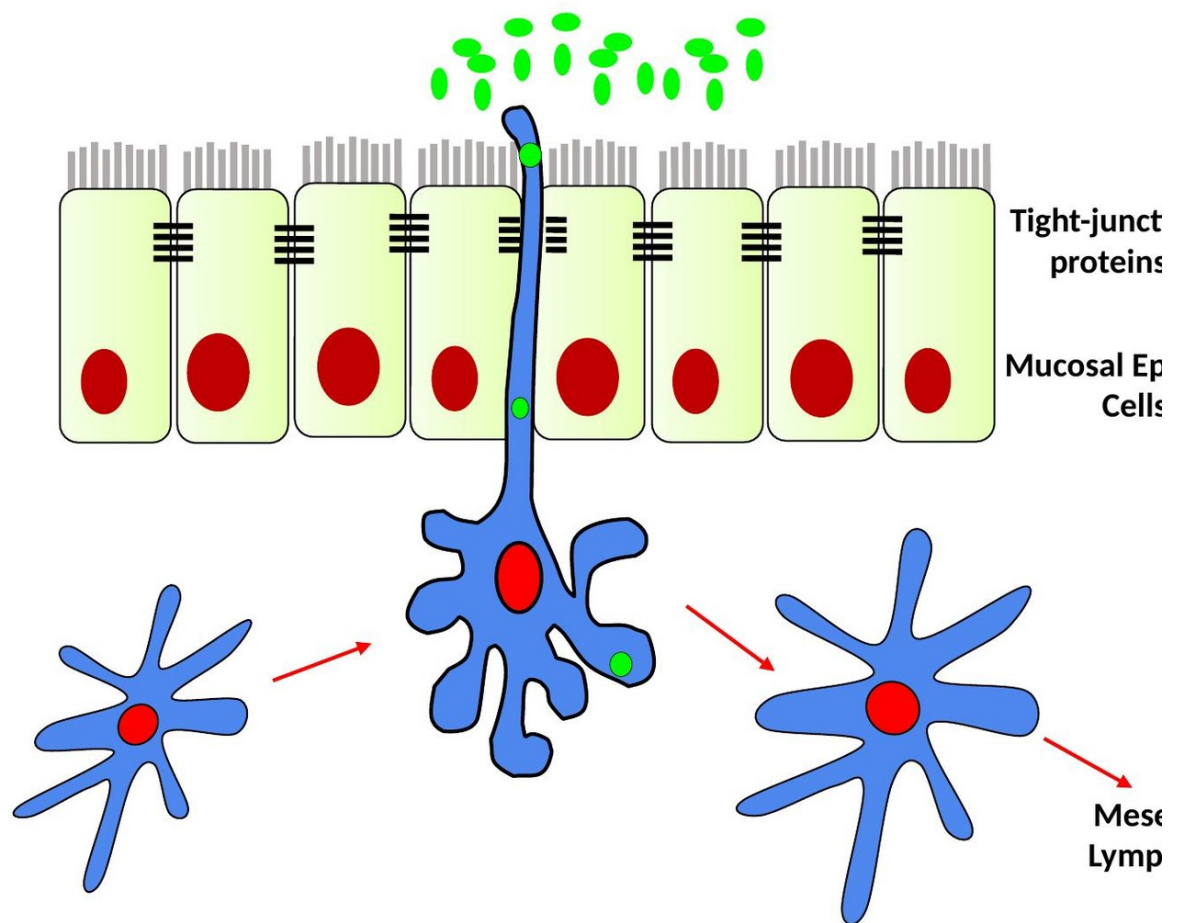
le Sensors”



at the gastrointestinal tract and relate to local function

e microbiome in maintaining systemic immune homeostasis

Figure L1.5 — Peyer's patch architecture: M cell in FAE samples luminal antigen → drops onto sub-epithelial dome (dendritic cells) → underlying organised follicle (B cells centre, naïve T cells periphery) → drains to mesenteric lymph node.



Cells of the gastrointestinal tract: Describe the cellular arrangement of tissues throughout the gastrointes

Figure L1.6 — Trans-epithelial dendritic cell antigen sampling. DC opens tight junctions and extends a dendrite into the lumen to sample bacteria directly without breaching the barrier; then migrates to the mesenteric lymph node.

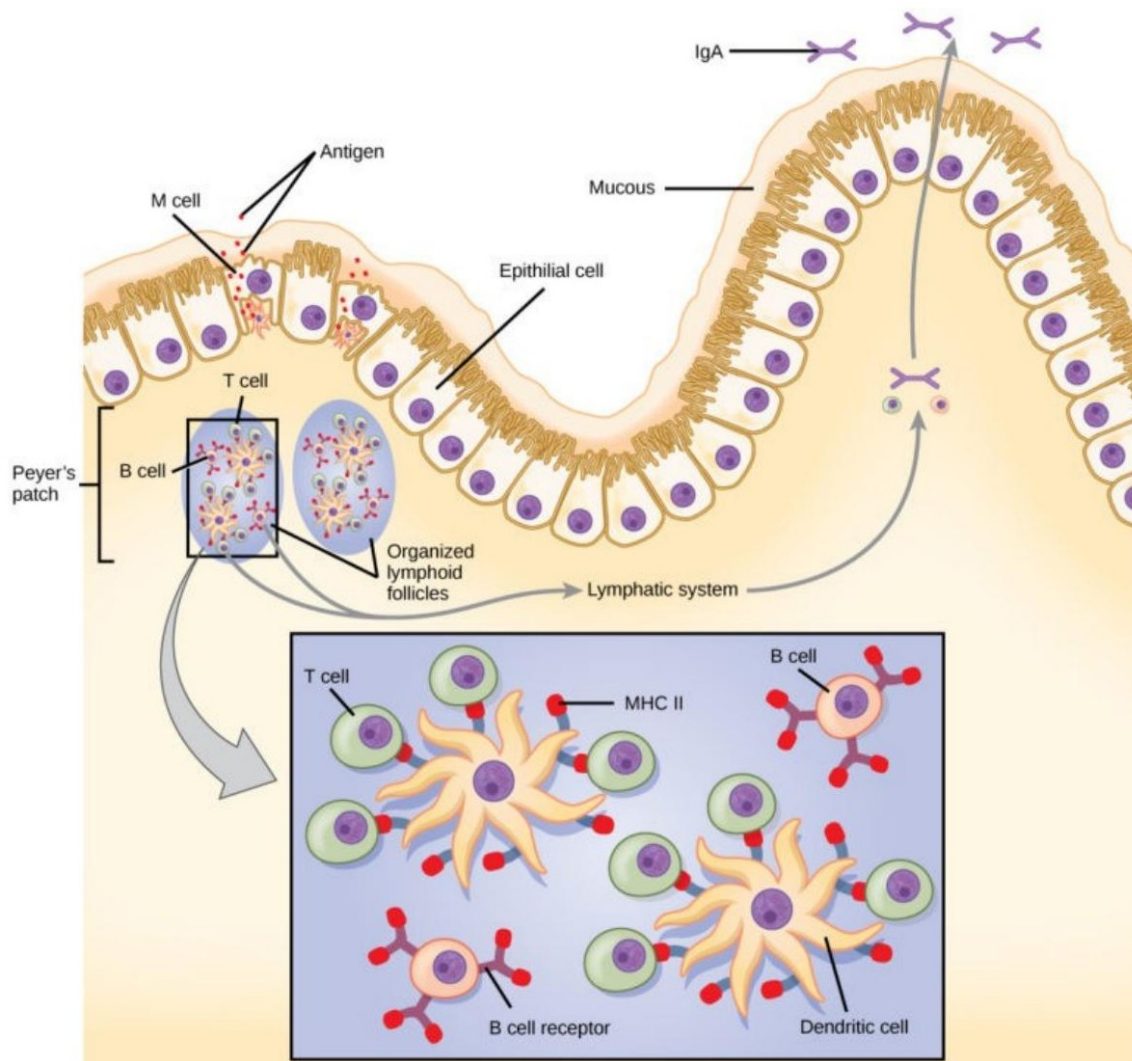
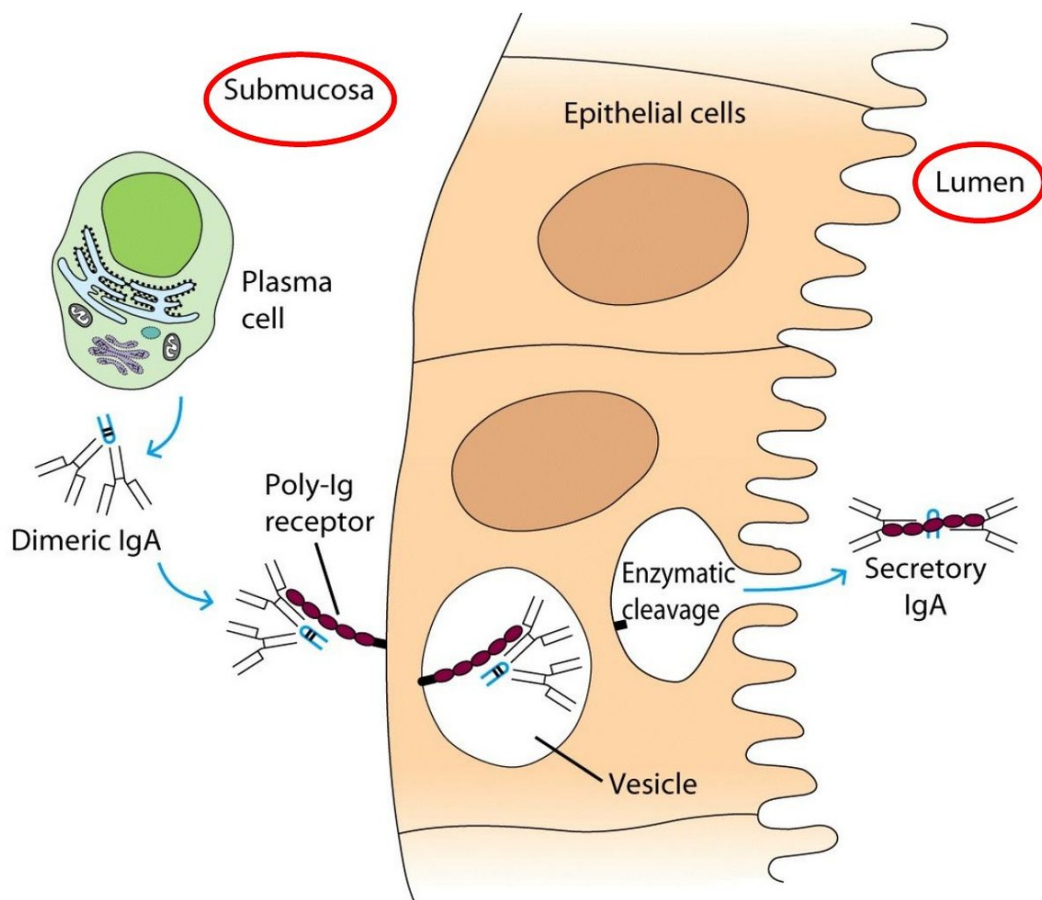


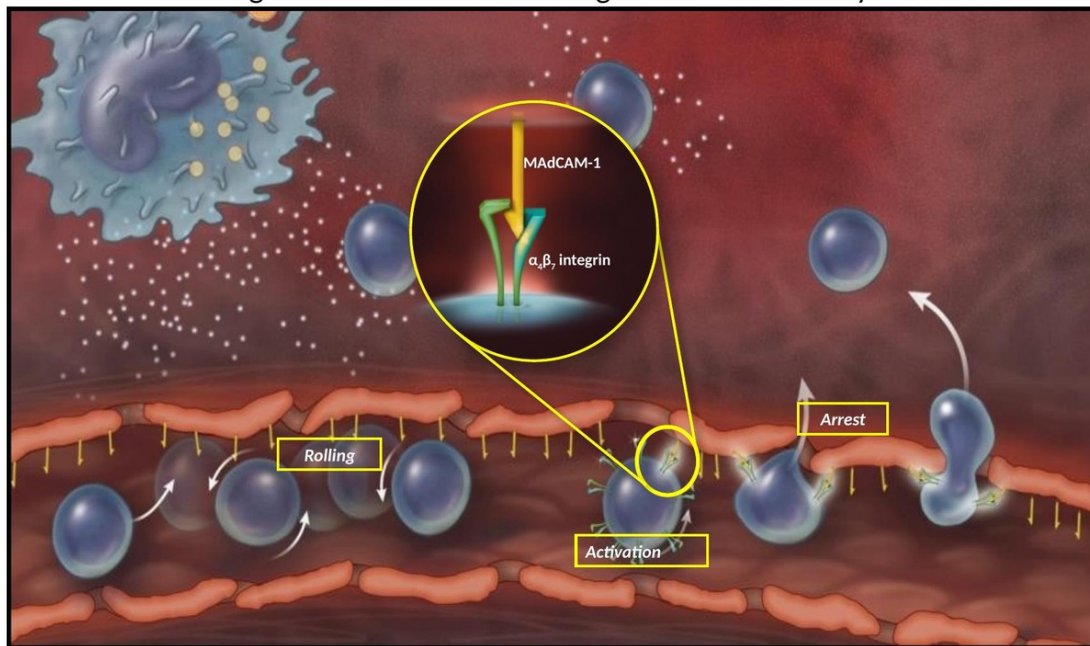
Figure L1.7 — Gut B-cell adaptive response: M cell uptake → Peyer's patch (T/B/DC interactions, MHC II antigen presentation) → IgA-secreting plasma cells in lamina propria → IgA secreted across epithelium into mucus.



Cells of the gastrointestinal tract: Describe the cellular arrangement of tissues throughout the gastrointes
Immunology of the gut: Outline the role of the gastrointestinal tract immunology and the microbiome in r

Figure L1.8 — Secretory IgA formation. Plasma cell secretes dimeric IgA → binds poly-Ig receptor on basolateral epithelium → transcytosis in vesicle → enzymatic cleavage at apical surface releases sIgA, with the receptor remnant retained as the secretory component.

Homing cascade directs circulating Naive T Cells to Peyer's Patches

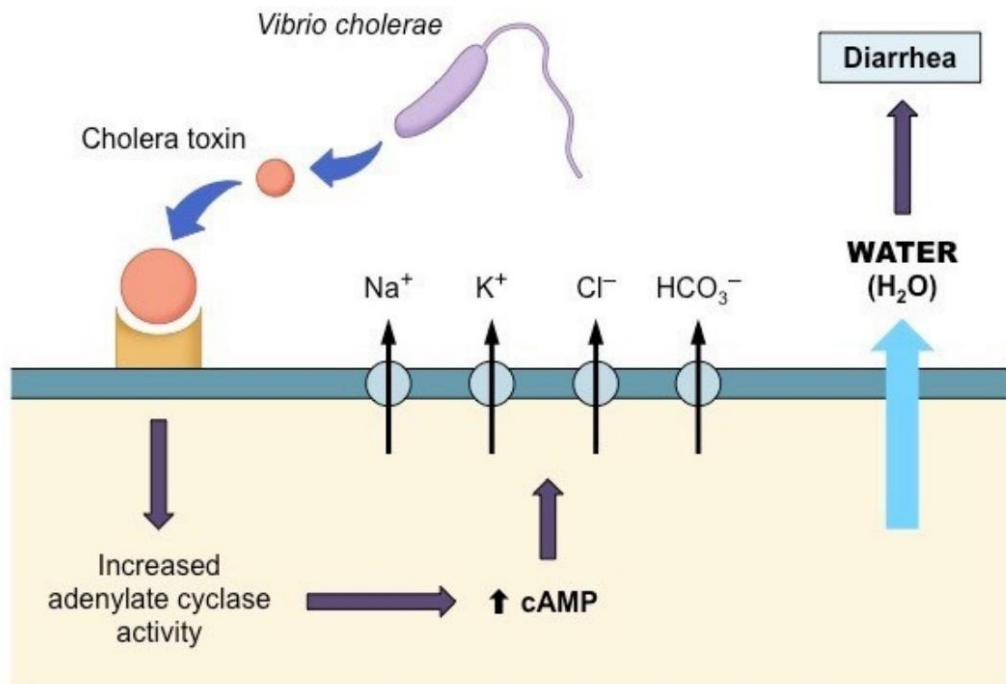


MAdCAM1: Mucosal Addressin Cell Adhesion Molecule 1; tissue specific

Describe the cellular arrangement of tissues throughout the gastrointestinal tract and relate to local function

Figure L1.9 — Gut homing cascade. T cell rolls on HEV → chemokine activation → $\alpha_4\beta_7$ integrin binds MAdCAM-1 (gut-specific addressin) → arrest → transmigration into lamina propria.

- Bacteria reaches small intestine → contact with epithelium & releases **cholera enterotoxin**



Intestinal disorders: Describe the clinical features and treatment options of small a

Figure L1.10 — Cholera toxin mechanism. Toxin enters enterocyte → activates adenylate cyclase → ↑ cAMP → CFTR pumps Cl^- into lumen → $\text{Na}^+ + \text{H}_2\text{O}$ follow → massive watery secretion (rice-water diarrhoea).

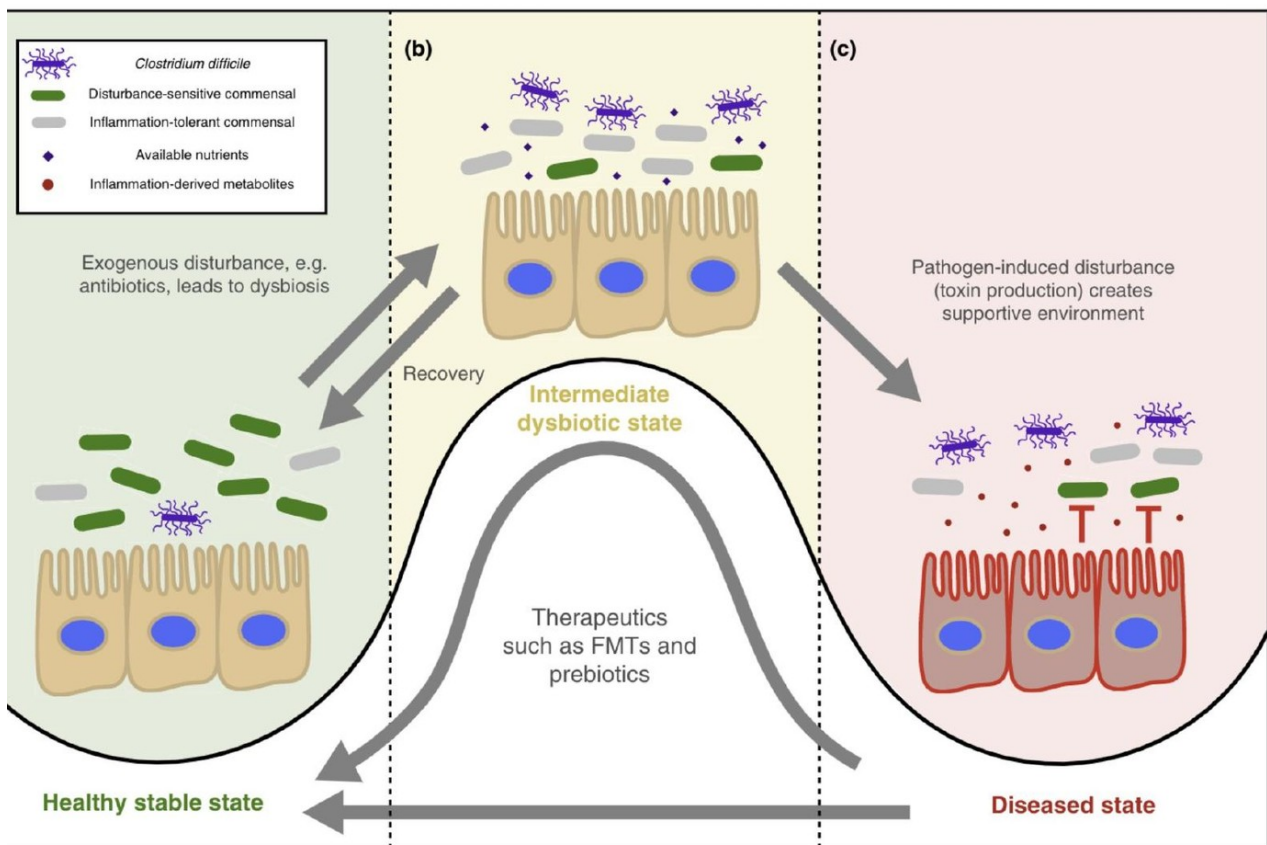


Figure L1.11 — *C. difficile* dysbiosis cycle. (a) Healthy stable state with commensals dominant → (b) antibiotic disturbance → intermediate dysbiotic state → (c) toxin-producing diseased state. Recovery via FMT/prebiotics.